

EEG Processing for Brain-Computer Interface using Wavelet-Decomposition Ant Colony Optimization and Machine Learning

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Abstract— The research in this paper leverages Electroencephalography (EEG) to develop a Brain-Computer Interface (BCI). By processing and analyzing EEG signals, the system translates brain activity into commands, enabling individuals with severe motor impairments to control devices and regain independence. The Berlin BCI Competition IV dataset is used in the study. EEG recordings of healthy subjects doing hand and foot motor imagery tasks are included in the collection. Discrete wavelet transformation (DWT) decomposes the EEG signals into sub-bands to obtain pertinent information. Feature selection is performed using the ant colony optimization algorithm (ACO) to reduce dimensionality and improve classification accuracy. An evaluation of three robust classifiers—support vector machine (SVM), weighted k-nearest neighbors (Weighted k-NN), and wide neural networks (WNN)—is conducted to ascertain their capability in Motor Imagery (MI) task classification. The results demonstrate that the SVM classifier in conjunction with DWT and ACO, achieves the highest average accuracy score of 85.36% while achieving a dimension reduction of 2.064-times. It makes it an appealing approach for the BCI systems.

Keywords— *Ant Colony Optimization, Brain-computer interface, Classification, Discrete Wavelet Transform, Evaluation measure, Electroencephalogram, Features selection, Motor imagery tasks, Machine learning*

I. INTRODUCTION

Brain-Computer Interfaces (BCIs) offer a promising avenue for individuals with motor impairments to interact with the world. These systems bypass conventional neuromuscular channels and allow direct communication between the brain and external equipment [1]. While invasive and semi-invasive BCI techniques exist, non-invasive methods, such as Electroencephalography (EEG), are particularly attractive due to their affordability, portability, and non-invasiveness [2], [3]. With the advancement of technology, the feasibility of developing the affordable and robust BCI systems is improving. This is the reason that the application of BCI systems is increasing in several applications.

A. Related work

In [4] a survey is presented, focusing on the integration of BCI solutions with augmented and virtual reality approaches. It is mentioned that such an approach can lead towards

experiencing the virtual societies, virtual offices, and even an experience of living in a virtual world. In [5], a survey is presented on the application of BCI solutions for the rehabilitation. In [6] a historical review of the BCI solutions is presented with present state and future directions. BCI applications have expanded to encompass a wide range of areas, including controlling computers, cursors, and peripheral devices [11, 12, 13]. As technology continues to evolve, BCI systems hold the potential to revolutionize the lives of individuals with disabilities, providing them with greater independence and quality of life.

The EEG allows a realisation of the BCIs. The major benefit is a non-invasive EEG measurement plus portability and affordability. Despite its advantages, EEG signals are susceptible to noise and artifacts arising from physiological and environmental factors. Effective signal processing techniques, including filtering and conditioning, are crucial to extract meaningful information from the raw EEG data [7]. Throughout numerous years, the domain of Brain-Computer Interfaces (BCI) has achieved extraordinary advancement. In 1973, Professor Jacques Vidal coined the term "brain-computer interface" and demonstrated how EEG signals enabled cursor movement through a two-dimensional maze [6]. P300 spellers, introduced by Farwell and Donchin in 1988, allowed individuals to select letters mentally by eliciting specific brain responses [8], [9]. Functional Magnetic Resonance Imaging (fMRI) has been used to explore brain activity during cognitive tasks, including real-time interaction between individuals [10]. In [11] the EEG signals were employed for the realisation of an online control an upper-limb assisting robot. In [12], the intentions, carried by the EEG signals, are decoded and used for controlling the movement of a cursor. The wireless wearable sensors are beneficial for the realization of unobstructive solutions. In this framework, in [13], a survey is presented on the wireless EEG recording methods and their application is the BCI solutions.

B. Contribution

Developing successful assistive and rehabilitative solutions may result from a non-invasive, EEG-based method for identifying motor imagery (MI) tasks. In this regard, a new hybridization of the ant colony optimization algorithm (ACO) and the discrete wavelet transform (DWT) is introduced in this study for efficiently extracting and

identifying the most pertinent MI categorization related features in the EEG signals. The categorization module receives the chosen feature set. To analyze this extracted feature set for an automatic categorization of the MI tasks, the effectiveness of three robust classifiers—the "support vector machine" (SVM), "weighted k-nearest neighbors" (Weighted k-NN), and wide neural networks (WNN)—is assessed. The performance is evaluated using a real EEG MI dataset. Additionally, multiple evaluation measures with cross-validation strategy are used to avoid any biasness in findings.

C. Paper organization

The remainder of the paper is arranged as follows: Section II summarizes the materials and methods used in this research. Section III details the evaluation metrics employed to assess the models. Simulation results and their discussion are available in Section IV. Finally, Section V concludes the paper and proposes future research directions.

II. MATERIALS AND METHODS

With the aid of Fig. 1, the general plan of the developed solution is illustrated. EEG data collection, pre-processing, feature extraction, dimension reduction through feature selection, and classification are essential stages. After acquiring and pre-processing EEG data to eliminate noise, pertinent features are extracted by decomposing the data using the DWT. The BCI system can then correctly interpret the MI tasks using machine learning techniques to categorize these features into various motor imagery tasks.

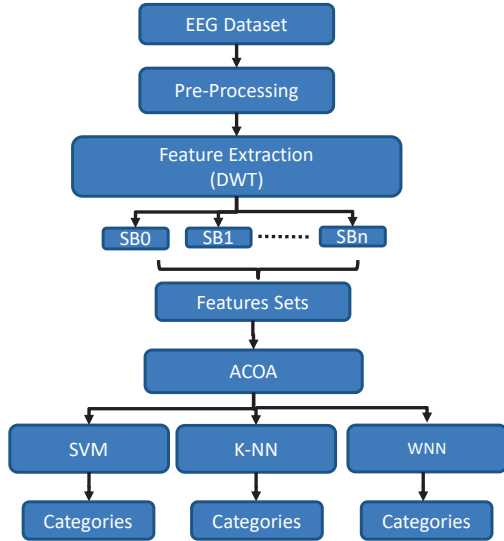


Fig. 1. The proposed scheme.

D. Dataset

This research utilizes the BCI Competition IVa dataset [14], collected from five healthy volunteers. Each participant performed 140 trials for two motor imagery tasks (hand and foot movements), resulting in 280 trials per participant. Each trial lasted 3.5 seconds. The dataset comprises training and testing sets of varying sizes for each task.

E. Pre-processing

The signal is pre-processed using a Discrete-Time Finite Impulse Response (FIR) filter. As shown in Equation (1) this

filter generates an output sequence by summing weighted input values [15].

$$y[n] = \sum_{i=0}^{N-1} b_i \cdot x[n-i] \quad (1)$$

Where, $x[n]$ is the input signal, $y[n]$ is the output signal, N is the filter order, b_i is the value of the impulse response at the i^{th} instant.

To isolate specific segments of the signal, a window function is applied. This technique sets values outside the desired window to zero, effectively focusing on the relevant portion of the signal and suppressing irrelevant information [16].

F. Discrete Wavelet Transform (DWT)

The windowed signal is further decomposed into sub-bands using the Wavelet Transform (WT). This mathematical transformation is defined in Equation (2) as follows [17]:

$$W_x^\psi(u, s) = \frac{1}{\sqrt{s}} \int_{-\infty}^{+\infty} x(t) \psi \left(\frac{t-u}{s} \right) dt. \quad (2)$$

Where, s represents the dilation parameter, $\psi(t)$ is the mother wavelet function, and u the translation parameter.

The DWT is used to get an in-depth sight from denoised EEG signals. It permits to extract valuable patterns and contents from the intended signals. In this study each instance is decomposed in five sub-bands using a fourth level of decomposition. The method is based on applying a digital filtering procedure with multiple levels. A cascaded assembly of the high-pass $h[n]$ and low-pass $g[n]$ filters processes the input EEG instance. Consequently, several sub-bands are produced. The low-frequency component is further broken down into other stages after the first stage's conclusion, giving the signal a hierarchical representation.

Daubechies wavelets are employed in this study, providing a robust and efficient approach to signal decomposition. The DWT generates approximation coefficients, a_m^i , which capture the overall signal trend, and detail coefficients, d_m^i , which represent the signal's fine-grained features at different frequency bands. Equations (3) and (4) mathematically describe the computation of these coefficients.

$$a_m^i = \sum_{k=1}^{K_g^i} x f_n^i \cdot g_{2n-k} \quad (3)$$

$$d_m^i = \sum_{k=1}^{K_g^i} x f_n^i \cdot h_{2n-k} \quad (4)$$

Where m represents the level of decomposition.

Each sub-band is then analyzed using various feature extraction techniques, including statistical measures (skewness, standard deviation, mean absolute value, energy, entropy), and time-domain features (root mean square (RMS), absolute peak-to-peak difference (APPD) and zero-crossing (ZC) rate). The computation of these time-domain features is detailed in Equations (5), (6), and (7), respectively.

$$RMS = \sqrt{\frac{1}{M} \sum_{k=1}^M IA(i, k)^2 \forall f_i}. \quad (5)$$

Where, M shows the number of samples of the windowed signal and f_i is the frequency of the i^{th} component.

$$APPD = |x_{max} - x_{min}|. \quad (6)$$

In the above equation, x_{max} and x_{min} indicate the maxima and minima of the segmented signal, respectively.

$$ZC = \begin{cases} 1, & \text{if, } (x_k > 0 \text{ AND } x_{k+1} < 0 \\ & \text{OR } (x_k < 0 \text{ AND } x_{k+1} > 0) \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

Dataset dimensionality can be reduced through feature selection. By reducing the number of features, we can build models more efficiently, with lower computational complexity, and improve the learning and generalization performance of machine learning algorithms.

G. Ant Colony Optimisation Algorithm (ACO)

Dimension reduction is used to get rid of the high dimensionality of the feature set. It not only diminishes the dataset size but also reduces the ambiguity among different classes and finally renders an enhanced post classification performance in terms of computational cost and categorization precision. However, while decreasing the number of attributes the vital information must be preserved. Means most of the variability of the original dataset must be retained in the selected features as feasible.

In this study the ‘‘ant colony optimization algorithm’’ (ACO) is employed for feature selection. It is an optimization algorithm inspired by nature and based on how ants forage. It is employed to resolve intricate optimization issues, especially those that entail identifying the most crucial individuals within a population [18]. ACO is based on stigmergy, a form of indirect communication where ants alter their local environment to guide other ants. A key aspect of ant behavior is their ability to find the shortest paths between their nest and food sources by following pheromone trails. Ants decide on a path based on the intensity of the pheromone, and over time, the pheromone evaporates, influencing the path choice. The flowchart for the ACO process is shown in Fig. 2.

H. Classification

The ACO algorithm selects 47 features from the original 97-dimensional feature collection. It produces a 2.06-times reduction in dimension. The DWT, ACOA, and classifiers are implemented using the MATLAB R2021a. The classification performance of three machine learning classifiers is compared. These are the SVM, Weighted k-NN, and WNN. A 10-fold cross-validation method is used to evaluate these classifiers' performance [19].

- *Support Vector Machine classifier (SVM)*

The SVM classifier employing kernel techniques represents a multifaceted supervised machine learning algorithm that can categorize linear and non-linear datasets. It operates by building ideal hyperplanes that optimize the margin between various classes—soft-margin SVMs trade off perfect separation for better generalization by permitting

some misclassification to increase flexibility. In a one-dimensional case, the SVM classifier can be visualized as a single point on a number line [20].

In two-dimensional datasets with linear separability, the support vector classifier manifests as the separating line while the soft margin emerges from distances to the support vectors. New data points are classified based on their position relative to this line. In three-dimensional space, a plane serves as the support vector classifier, and points are classified according to their position above or below the plane.

The support vector classifier functions as a hyperplane in reduced dimensional space relative to the data it separates. For example, a one-dimensional hyperplane (a point) is used for one-dimensional data, a two-dimensional hyperplane (a line) for two-dimensional data, and so on. Equation (8) mathematically describes the process of constructing the separating hyperplane.

$$h(x) = \text{sign}(\omega \cdot x^T + b). \quad (8)$$

Where x is a sample vector $x = [x_1, x_2 \dots x_p]$ with p attributes, $\omega = [\omega_1, \omega_2 \dots \omega_p]$ is the weights vector, and b is a scalar bias.

In this study the SVM is employed with polynomial kernel and the complexity parameter value of 50 is found to be the most suitable.

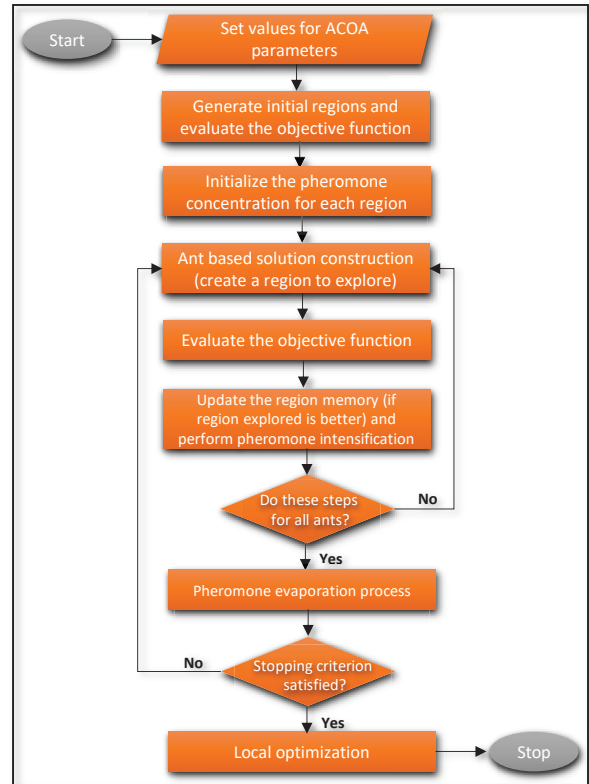


Fig. 2. The flowchart of ACOA.

- *k-Nearest Neighbors (k-NN)*

The k-NN is another straightforward classification technique. According to this technique, comparable data points are probably part of the same class. In order to classify

a new data point, k-NN uses a distance metric to identify its k-nearest neighbors and assigns the most common class label to the new point among these neighbors. For instance, the new point is given the label of its closest neighbor if $k=1$. If $k=15$, the majority vote among the 15 nearest neighbors determines the class label [21].

Letting v_j represent the sample and $\langle v_j, l_j \rangle$ denote a tuple of a training sample and its label, $l_j \in [1, C]$ where C is a number of classes. Mathematically, given a test sample z , the nearest neighbor j can be determined using the formula in equation (9):

$$\operatorname{argmin} \operatorname{dist}(d_j, t) \forall j = 1 \dots N. \quad (9)$$

In this study the k-NN is used with the Euclidean distance and value of $k=11$ is found to be the most suitable.

- *Wide Neural Network (WNN)*

The WNNs are machine learning models with many neurons in their hidden layers, unlike deep learning models that prioritize having more layers. The design of these models takes inspiration from the intricate connections between neurons in the human brain. An artificial wavelet neural network comprises multiple interconnected nodes arranged in distinct layers, including an input layer, followed by one or more hidden layers, concluding with an output layer. The input layer receives the data, while the hidden layers use activation functions like Softplus, ReLU, or Sigmoid to handle it. By adding non-linearity, these functions enable the network to identify intricate connections in the data. The hidden layers polish the input through a series of modifications, allowing the model to identify significant patterns. Finally, the output layer delivers the classification result. This layered approach makes WNNs highly effective for solving diverse machine learning tasks [22].

In this study the WNN is used with two hidden layers.

III. EVALUATION MEASURES

Performance metrics derive from each classifier's confusion matrix values, include the "True Positives" (TP), "True Negatives" (TN), "False Negatives" (FN), and "False Positives" (FP). This procedure for two-class classification is depicted in Fig. 3. A model achieves excellence by maximizing true positives and negatives while minimizing false positives and false negatives to boost the performance metrics.

		Actual Values	
Predicted Values		Positive	Negative
	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

Fig. 3. Confusion matrix for the case of a two-class classification problem.

To achieve dependable performance benchmarks while delivering consistent real-world prediction accuracy, classification models require extensive evaluation processes. Performing numerous evaluations boosts analytical reliability. A thorough evaluation of the chosen model necessitates

testing in both training environments and unobserved testing conditions [17]. The evaluation process requires an intricate examination of the model's outputs alongside their corresponding actual ground truth values. In practical scenarios the access to datasets especially those with labels remains frequently restricted. To tackle this problem, the dataset undergoes meticulous partitioning into multiple subsets through cross-validation. Evaluating the proposed ensemble learning classifiers' performance utilizes the 10-fold cross-validation method.

The research investigation implemented a variety of assessment tools which encompassed "Accuracy" (Acc), "F-measure (F1)", "Specificity", "Kappa Index", and "Area Under the Curve" (AUC). The determination of these metrics occurs through experimental generation of confusion matrices. Binary classification tasks utilize a tabular confusion matrix to encapsulate model performance metrics.

The results of the evaluation metrics under consideration are used to determine which AI model is best suited for the intended use. The process assists in selecting the optimal model among multiple alternatives. Various evaluation metrics enable decision-makers to gather more comprehensive information for their choices. This study employs a range of metrics, including accuracy, precision, recall, specificity, F-measure, and the area under the ROC curve (AUC). The confusion matrices, which were discovered through experimentation, are used to measure these.

The following equations show how to calculate the evaluation metrics that are considered.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}. \quad (10)$$

$$\text{Precision} = \frac{TP}{TP+FP}. \quad (11)$$

$$\text{Recall} = \frac{TP}{TP+FN}. \quad (12)$$

$$\text{Specificity} = \frac{TN}{TN+FP}. \quad (13)$$

$$F - \text{measure} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}. \quad (14)$$

AUC visually represents a classifier's categorization capabilities [17]. The True Positive Rate (TPR) is represented by the y-axis, and the False Positive Rate (FPR) by the x-axis. Equations (15) and (16) define the TPR and FPR, respectively.

$$\text{TPR} = \frac{TP}{TP+FN}. \quad (15)$$

$$\text{FPR} = \frac{FP}{FP+TN}. \quad (16)$$

IV. RESULTS AND DISCUSSION

The actual EEG-based MI tasks dataset is used to assess the performance of the developed method. This dataset initially contains 840 cases in total for the two classes. Each class has equal representation. Means, 420 instances per category. Each segment is pre-processed for noise removal and onward decomposed using the DWT. The DWT renders 97 features per instance. After the application of ACOA, 47 features are selected. It resulted in a dimension reduction of

2.064-times. It promises to significantly lower the computational cost and latency associated with data storage, transmission, and post-classification. The categorization stage receives the prepared feature set. The processing performance of SVM, Weighted k-NN, and WNN classifiers is examined to classify the two classes of MI tasks automatically. The 10-CV strategy is used to test the classification performance.

The performance measures differ across the classifiers considered in this study: SVM, WNN, and Weighted k-NN. The results for each classifier are respectively presented in Tables I-III. In these Tables, C1 and C2 represent Class-1 (Hand) and Class-2 (Foot), respectively.

TABLE I. DIFFERENT EVALUATION MEASURES FOR THE CUBIC SVM CLASSIFIER

Performance Metric	Hand Class (C1)	Foot Class (C2)	Average Performance
Accuracy	0.854	0.854	0.854
Precision	0.849	0.858	0.854
Recall	0.859	0.848	0.854
Specificity	0.848	0.859	0.854
F1-score	0.854	0.853	0.854
AUC	0.920	0.920	0.920

TABLE II. DIFFERENT EVALUATION MEASURES FOR THE WEIGHTED K-NN CLASSIFIER

Performance Metric	Hand Class (C1)	Foot Class (C2)	Average Performance
Accuracy	0.848	0.848	0.848
Precision	0.810	0.883	0.847
Recall	0.895	0.790	0.843
Specificity	0.790	0.895	0.843
F1-score	0.851	0.834	0.842
AUC	0.920	0.920	0.920

TABLE III. DIFFERENT EVALUATION MEASURES FOR THE WNN CLASSIFIER

Performance Metric	Hand Class (C1)	Foot Class (C2)	Average Performance
Accuracy	0.783	0.783	0.783
Precision	0.785	0.782	0.783
Recall	0.781	0.786	0.783
Specificity	0.786	0.781	0.783
F1-score	0.783	0.784	0.783
AUC	0.860	0.860	0.860

A classification evaluations performance comparison is also graphically depicted among the three classifiers in Fig. 4.

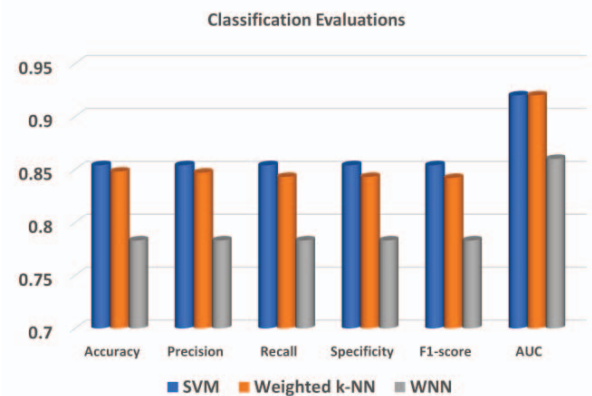


Fig. 4. The classification evaluations comparison between the three classifiers.

The evaluation of the three classifiers, SVM, Weighted k-NN, and WNN, reveals clear differences in performance across six metrics: accuracy, precision, recall, specificity, F1-score, and AUC.

The SVM emerged as the best-performing classifier overall. It achieved the highest average accuracy (85.4%), indicating reliable overall predictions. Its precision (85.4%) and recall (85.4%) were balanced across both classes, showing consistent detection of Class 1 (Hand) and Class 2 (Foot). The F1-score (85.4%) also highlighted its ability to balance precision and recall effectively. With an AUC of 92%, Cubic SVM demonstrated robust class separation, making it the most suitable option for general use.

The Weighted k-NN followed closely, achieving an average accuracy of 84.8%. However, it exhibited noticeable variability between the classes. For Class 2 (Foot), it achieved high precision (88.3%) and specificity (89.5%), suggesting strong performance in correctly identifying this class. In contrast, its precision (81%) and specificity (79%) for Class 1 (Hand) were lower, indicating a potential bias toward Class 2. The recall for Class 1 was high (89.5%), but for Class 2, it was relatively low (79%). Weighted k-NN shared the same AUC (92%) as Cubic SVM, showing strong discriminatory ability but slightly less balanced performance across the metrics. On the other hand, WNN underperformed compared to the other two classifiers. Its average accuracy (78.3%), precision (78.3%), recall (78.3%), and F1-score (78.3%) were consistently lower, reflecting weaker predictions across both classes. While its precision and recall were balanced, they were less effective overall. The AUC (86%) was also the lowest, indicating poorer class separation. This suggests that the WNN may require optimization or architectural adjustments to perform competitively.

The system's compression, resource utilization, and power consumption efficiency can be increased tactically using the event-driven and adaptive rate pre-processing and recognition approaches [15]. It is advantageous for embedded and real-time implementation. In order to improve the system's computational and classification performance, other

optimizers-based feature selection and dimension reduction may be helpful [23], [24]. It might be possible to look into the viability of using these tools in the suggested approach.

V. CONCLUSION

For an automated categorization of the Motor Imagery (MI) tasks, this paper proposes an effective hybridization of digital filtering-based denoising, "discrete wavelet transform" (DWT)-based feature extraction, "ant colony optimization algorithm" (ACOA)-based feature selection, and machine learning-based classification. The non-invasive EEG waves are processed using the developed approach for the desired classification. The prepared features set categorizes the MI tasks using three machine learning classifiers: Weighted k-NN, SVM, and WNN. The results showed that the developed approach achieved the most significant average classification accuracy score of 85.40% using the SVM classifier. This superior performance is attributed to the enhanced feature extraction and selection capabilities of the DWT and ACOA, leading to improved classification accuracy. In summary, SVM stands out in categorization performance, making it the preferred choice for this task. Weighted k-NN performs well but shows a bias toward Class 2, which might be advantageous in specific applications where classification precision for Class 2 is critical. WNN requires significant improvements to be viable for this problem.

A new MI dataset can be collected to validate the suggested approach further, increasing the number of participants and trials. The system's overall performance and applicability would be improved by validating its performance for further potential datasets. In addition, future research will explore the possibility of incorporating other promising machine learning techniques, such as decision trees, random forests, and ensemble learning methods.

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